

Dynamic Partial Cointegration & Neural Kalman Filtering

for Statistical-Arbitrage Pairs Trading

Aryan Deo (230213)

November 2026

Undergraduate Project, MTH497 · Supervisor: Prof. Amit Mita
Department of Mathematics & Statistics, IIT Kanpur

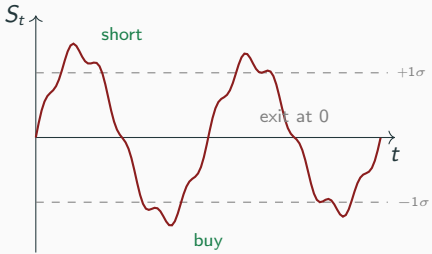
Pairs trading: the bet

Two related assets tend to move together.

- Build a hedged spread: $S_t = X_{2,t} - \beta X_{1,t}$
- When S_t is unusually wide, short it; when unusually narrow, buy it.
- Profit when the spread reverts. Market direction is hedged out.

The classical recipe (Engle–Granger) assumes:

1. one hedge ratio β, estimated once by OLS;
2. the whole spread is mean-reverting.



Problem statement: both assumptions break in practice

Failure 1 — the hedge ratio drifts

Capital structure, leverage and business mix change over years.
A fixed β slowly turns a hedged trade into a directional one.

Failure 2 — the spread has a permanent part

Spread = temporary M_t + permanent R_t.
Betting that R_t will "come back" is betting against a random walk.

Example: **NKE–AES** with a static hedge, 2017–2021 out of sample:
total return -64.4%, max drawdown -69.5%.

Goal: a filter that *tracks* β_t, *separates* M_t from R_t, trades only M_t — and learns to do this robustly on real market data.

Research questions

1. Can one state-space filter track a time-varying hedge ratio β_t *and* split the spread into tradeable and non-tradeable parts?

2. Can a small neural network learn the Kalman gain — trained directly on a cost-aware Sharpe objective — and beat the analytical gain out of sample?

3. Which statistics, computed before trading, separate pairs that will make money from pairs that will not?

Inspiration: three papers, one framework

Source	Key idea	What we take
Clegg & Krauss (2018)	Partial cointegration: spread $=$ $M_t + R_t$	Trade only M_t; variance-ratio screen
Yu (2022)	Kalman filter for a dynamic hedge ratio	β_t as a latent random walk
Revach et al. (2022) Milstein et al. (2024)	KalmanNet: GRU learns the gain; applied to pairs trading	Learned gain, two-stage training

What is new here: all three combined in a single filter, trained with a differentiable Sharpe loss with transaction costs inside it, plus a pre-trade screen and an online risk gate.

The Dynamic-PCI state-space model

$$X_{2,t} = \underbrace{\beta_t X_{1,t}}_{\text{hedge leg}} + \underbrace{M_t}_{\text{temporary}} + \underbrace{R_t}_{\text{permanent}} + \underbrace{v_t}_{\text{noise}}, \qquad v_t \sim \mathcal{N}(0, \sigma^2_{\text{obs}})$$

State equation: $z_t = F z_{t-1} + w_t, \;\; w_t \sim \mathcal{N}(0, Q)$

$$\begin{bmatrix} \beta_t \\ M_t \\ R_t \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \rho & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_{t-1} \\ M_{t-1} \\ R_{t-1} \end{bmatrix} + w_t$$

$$Q = \operatorname{diag}(q_\beta^2,\ \sigma_M^2,\ \sigma_R^2)$$

Observation equation (scalar):

$$X_{2,t} = \underbrace{\begin{bmatrix} X_{1,t} & 1 & 1 \end{bmatrix}}_{H_t} z_t + v_t$$

- $X_{1,t}$ is observed, so it enters via H_t, not the state
- Linear in z_t but time-varying $\Rightarrow$ standard Kalman filter
- Prices normalised to 1.0 on day one: all terms $O(1)$

What each state means

- β_t: hedge ratio, slow random walk ($q_\beta \ll \sigma_M$, so it absorbs only long-run drift)
- $M_t = \rho M_{t-1} + \varepsilon_t^M$, $|\rho| < 1$: mean-reverting part — the only thing traded
- $R_t = R_{t-1} + \varepsilon_t^R$: permanent drift — never traded

Parameters (MLE on formation window):
$\theta = (\beta_0, \rho, \sigma_M, \sigma_R, q_\beta)$

Nested special cases

$q_\beta = 0$	static PCI (Clegg & Krauss)
$\sigma_R = 0$	classical cointegration
$\rho \to 1$ or $\sigma_M = 0$	no tradeable spread

Kalman filter and the likelihood

Kalman filter — for fixed θ, compute $\hat{z}_{t|t} = \mathbb{E}[z_t \mid X_{1:t}]$ recursively:

Predict

$$\hat{z}_{t|t-1} = F\hat{z}_{t-1|t-1}, \qquad P_{t|t-1} = FP_{t-1|t-1}F^\top + Q$$

Innovation (one-step forecast error)

$$e_t = X_{2,t} - H_t\hat{z}_{t|t-1}, \qquad S_t = H_tP_{t|t-1}H_t^\top + \sigma^2_{\text{obs}}$$

Gain and update

$$K_t = P_{t|t-1}H_t^\top S_t^{-1}$$

$$\hat{z}_{t|t} = \hat{z}_{t|t-1} + K_te_t, \qquad P_{t|t} = (I - K_tH_t)P_{t|t-1}$$

- S_t is a scalar: the gain needs a division, not a matrix inverse
- K_t decides how much of the surprise e_t goes to β_t, M_t or R_t

Where the likelihood comes from

1. Chain rule — joint density = product of one-step forecasts:

$$p(X_{2,1:T} \mid \theta) = \textstyle\prod_t p\big(X_{2,t} \mid X_{2,1:t-1}; \theta\big)$$

2. Gaussian forecasts — $z_t \mid \text{past} \sim \mathcal{N}(\hat{z}_{t|t-1}, P_{t|t-1})$, $X_{2,t}$ linear in z_t:

$$X_{2,t} \mid \text{past} \;\sim\; \mathcal{N}\big(H_t\hat{z}_{t|t-1},\; S_t\big)$$

3. Take logs — prediction-error decomposition:

$$\boxed{\ell(\theta) = -\tfrac{1}{2}\sum_{t=1}^{T}\Big[\ln(2\pi S_t) + \frac{e_t^2}{S_t}\Big]}$$

- $\ln S_t$: penalises uncertainty; e_t^2/S_t: penalises surprise
- One filter pass gives $\ell(\theta)$ in $O(T)$; maximise by L-BFGS-B

Reading the fitted pair

With $\hat{\theta}$ from the MLE, difference the two spread parts:

$$\mathrm{Var}(\Delta M_t) = \frac{2\sigma_M^2}{1+\rho}, \qquad \mathrm{Var}(\Delta R_t) = \sigma_R^2$$

Share of spread variance that is mean-reverting:

$$R_{MR}^2 = \frac{2\sigma_M^2}{2\sigma_M^2 + (1+\rho)\sigma_R^2}$$

Since $\mathbb{E}[M_{t+h} \mid M_t] = \rho^h M_t$, the half-life is

$$\mathrm{HL} = -\frac{\ln 2}{\ln|\rho|} \text{ days}$$

How to read them

- $R_{MR}^2 \to 1$: spread is mostly temporary — highly tradeable
- $R_{MR}^2 \to 0$: permanent drift dominates — avoid
- Short HL: reversion happens fast enough to be paid within the trading horizon

These two numbers, computed on formation data only, drive the pre-trade screen (Result 3).

Trading strategy and signal

1. Signal — built only from the filtered temporary part $\hat{M}_{t|t}$, never from $\hat{R}_t$:

$$z_t = \frac{\hat{M}_{t|t} - \mu_M^{(80)}}{\sigma_M^{(80)}}$$

$\mu_M^{(80)}$, $\sigma_M^{(80)}$: rolling 80-day mean and std. of $\hat{M}$.

2. Rules (evaluated daily)

Condition	Action	Legs
$z_t \le -1$	long spread	buy X_2, short $\hat{\beta}_t X_1$
$z_t \ge +1$	short spread	short X_2, buy $\hat{\beta}_t X_1$
z_t crosses 0	exit	close both legs
loss $\le -10\%$	stop-loss	close both legs

3. Position sizing — legs in the hedge ratio, one unit of gross capital:

$$w_2 = \frac{1}{1 + |\hat{\beta}_t|}, \qquad w_1 = \frac{\hat{\beta}_t}{1 + |\hat{\beta}_t|}$$

e.g. $\hat{\beta}_t = 1.5$: 40% long X_2, 60% short X_1. Rebalanced daily as $\hat{\beta}_t$ moves.

4. Net daily return ($\mathrm{pos}_t \in \{-1, 0, +1\}$)

$$r_t = \mathrm{pos}_{t-1}\left(w_2 \frac{\Delta X_{2,t}}{X_{2,t-1}} - w_1 \frac{\Delta X_{1,t}}{X_{1,t-1}}\right) - \underbrace{2c \cdot \mathbf{1}\{\mathrm{pos}_t \neq \mathrm{pos}_{t-1}\}}_{\text{transaction costs}}$$

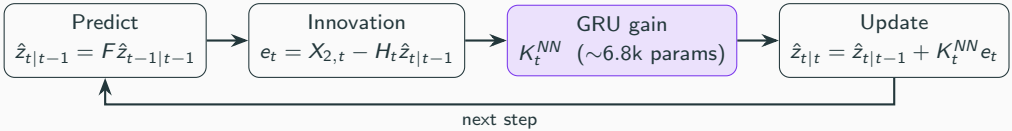
$c = 10$ bps per half-turn on each leg. The same rules and costs are used for all four models, so differences come from the filter alone.

Why learn the Kalman gain?

The analytical gain $K_t = P_{t|t-1} H_t^\top S_t^{-1}$ is optimal only if the model is exactly right. On market data it is not:

- returns are heavy-tailed $\Rightarrow$ Gaussian gains over-react to outliers;
- noise covariances (Q, R) are estimated with error;
- β_t is not truly a random walk.

KalmanNet idea: keep the model-based *predict* step, replace only K_t by a GRU.



The output stays an interpretable filter: we still read off $\hat{\beta}_t, \hat{M}_t, \hat{R}_t$.

Training: from tracking to trading

Stage 1 — MSE pre-training (15 epochs)

Minimise one-step prediction error of $X_{2,t}$. Anchors the gain near the analytical filter.

Stage 2 — Sharpe fine-tuning (15 epochs)

Maximise the Sharpe ratio of the strategy itself, net of costs.

Training profit alone without Stage 1 is known to diverge (Milstein et al.).

Hard entry rules have no gradient, so use a soft position:

$$\widetilde{\mathrm{pos}}_t = -\tanh\left(z_t/\tau\right)$$

$$R_t^{\mathrm{strat}} = \widetilde{\mathrm{pos}}_{t-1}\, r_t^{\mathrm{pair}} \;-\; c\,|\Delta\widetilde{\mathrm{pos}}_t|$$

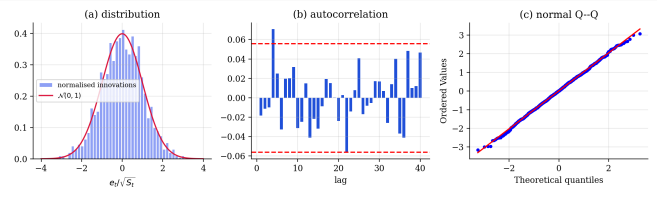
$$\mathcal{L}_{\mathrm{Sharpe}} = -\sqrt{252}\,\frac{\mathbb{E}[R_t^{\mathrm{strat}}]}{\sqrt{\mathrm{Var}(R_t^{\mathrm{strat}}) + 10^{-8}}}$$

Gradients flow through the position, the filter update and the GRU (BPTT). The cost term directly penalises turnover.

Sanity check on synthetic data

Simulated pair with known β_t, M_t, R_t ($n = 1250$, $\beta_0 = 1.5$, $\rho = 0.9$):

- β_0 and ρ recovered within $\approx 1\%$
- $\hat{\beta}_t$ tracks the true path: $r = 0.858$
- Innovations at true parameters are white & normal: Ljung–Box $p = 0.52$, KS $p = 0.69$, Shapiro–Wilk $p = 0.75$



Confirms the filter and likelihood are correct before touching real data.

Experimental setup

Data: 14 pairs, daily prices

- single stocks, sector & country ETFs, commodity/energy
- several chosen as deliberate stress tests

Strict time split

- Formation (fit + train): 2009–2017, 2,014 days
- Trading (out of sample): 2017–2021, 1,007 days, incl. COVID crash

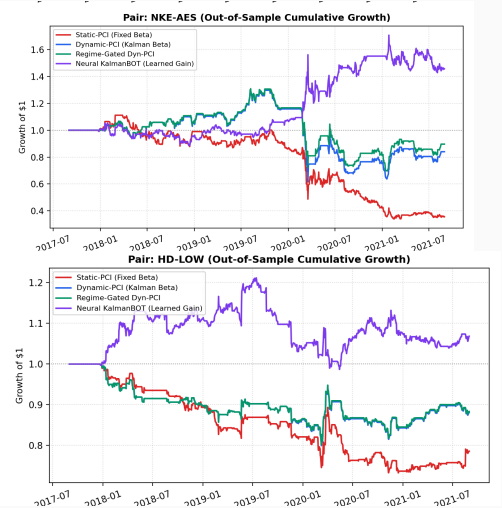
Four models (same data, signal, costs):

Model	What changes
Static-PCI	fixed β (baseline)
Dynamic-PCI	+ random-walk β_t
Regime-Gated	+ online risk gate
Neural KalmanBOT	+ learned gain

Result 1: losses from a static hedge are reversed

Pair	Model	Return	MaxDD
NKE–AES	Static	−64.4%	−69.5%
	Neural	+45.7%	−23.0%
HD–LOW	Static	−21.4%	−26.7%
	Neural	+6.8%	−18.6%
KO–PEP	Static	−9.1%	−23.2%
	Neural	+3.9%	−10.2%

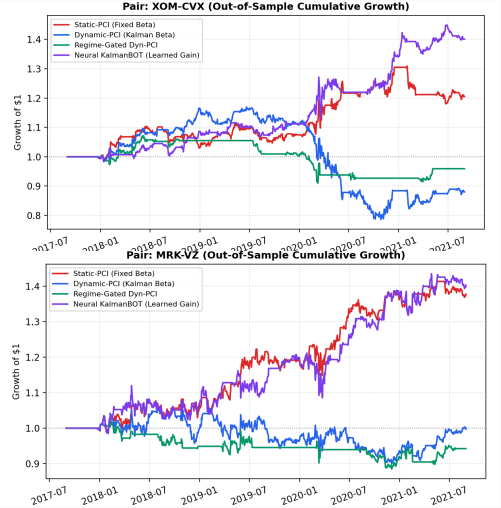
Where the fixed hedge drifts, the learned gain is positive on all three pairs.



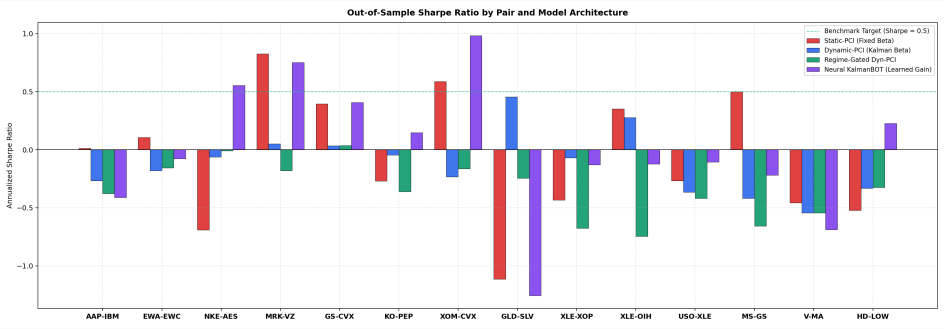
Result 2: where the static model works, Neural matches or beats it

Pair	Model	Return	Sharpe
XOM–CVX	Static	+20.5%	0.59
	Neural	+40.1%	**0.98**
MRK–VZ	Static	+37.7%	0.82
	Neural	+40.2%	0.75
GS–CVX	Static	+19.6%	0.40
	Neural	+32.4%	0.41

XOM–CVX Sharpe of 0.98 is the best in the study.



Across all 14 pairs



Model	Mean ret.	Mean MaxDD	Worst MaxDD	Pairs > 0
Static-PCI	-4.5%	-24.7%	-69.5%	6/14
Neural KalmanBOT	$+3.9\%$	-22.0%	-47.2%	6/14

Best mean return and smallest drawdowns of all four models.

Result 3: a screen that respects partial cointegration

Observation: the legacy ADF test on the raw spread contradicts PCI — if $\sigma_R^2 > 0$ the raw spread *has* a unit root, so good pairs get rejected.

Calibrated screen: $R^2_{MR} \geq 50\%$ and (raw ADF $p \leq 0.05$ or ADF on $\hat{M}_t$ $p \leq 0.01$)

Screen	Correct calls (of 14 pairs)
Legacy strict (Engle–Granger style)	10/14
Calibrated PCI	**12/14**

Recovers MRK–VZ, XOM–CVX and XLE–OIH, which the legacy screen wrongly rejected.

Caveat: thresholds were set after seeing outcomes — needs walk-forward validation.

Result 4: an online gate for capital preservation

A pair can break *after* it passes the screen. The gate flattens positions and blocks entries when

- spread volatility surges: $\sigma_{20d}/\sigma_{\text{hist}} > 3$, or
- mean reversion stalls: rolling 60-day half-life > 126 days.

Pair	Static MaxDD	Gated MaxDD	Reduction
GLD–SLV	−25.5%	−6.5%	**74%**
XLE–XOP	−20.4%	−0.6%	**97%**

Lower max drawdown than Dynamic-PCI on 9 of 14 pairs.

Failure: the gate is too conservative — it produced no positive return on any pair. Useful as a circuit breaker, not as a strategy.

Conclusion and next steps

What we showed

1. One filter tracks β_t and separates M_t from R_t; validated on synthetic data.
2. A learned, Sharpe-trained gain turns large static-hedge losses into gains (NKE–AES: $-64.4\% \rightarrow +45.7\%$) and has the best mean return and drawdowns.
3. A PCI-consistent screen classifies 12/14 pairs correctly.
4. An online gate cuts drawdowns by up to 97%.

To confirm it

- Walk-forward engine: 2-yr formation, 6-month trading, re-estimate continuously
- Multi-seed neural ensembles, formal significance tests
- Soft regime sizing (HMM) instead of an on/off gate
- Log-price (EKF/UKF) and multi-asset baskets

Thank you

References

1. M. Clegg and C. Krauss, "Pairs trading with partial cointegration," *Quantitative Finance*, 18(1), 2018.
2. J. Yu, "Cointegration approach for pair trading based on the Kalman filter," *BADS*, Atlantis Press, 2022.
3. G. Revach et al., "KalmanNet: Neural network aided Kalman filtering for partially known dynamics," *IEEE Trans. Signal Process.*, 70, 2022.
4. A. Milstein et al., "Neural augmented Kalman filtering with Bollinger bands for pairs trading," *IEEE Trans. Signal Process.*, 72, 2024.
5. R. F. Engle and C. W. J. Granger, "Co-integration and error correction," *Econometrica*, 55(2), 1987.
6. J. Durbin and S. J. Koopman, *Time Series Analysis by State Space Methods*, 2nd ed., OUP, 2012.